



R. J. Reynolds Industries, Inc.

Subject Side Stream apparatus and
workup -

Project No. 7607

Date 21 June 82

341101

In plumbing the Harrow stirred trap (HST) into the coolant and smoke collection "channels" of the Bagnall 20/18 smoking machine, the following was discovered. It should be noted that the Harrow stirred trap is used to collect the side stream smoke only, and that the 20/18 has a special "Head" attachment for the collection of side stream (SS) smoke.

The HST trap was connected to a LTE-9 Methlab cooler. The temperature of the coolant (in this case methyl alcohol) was -80°C . The reservoir of the HST is welded at the bottom and fitted with an "O" ring at the top. The original "O" ring was composed of Butyl rubber and it was found that the ring would not withstand the temperature of the coolant and would, over time, become too soft and the trap would leak. Upon inspection with a vision microscope and no further examination the leaking has been noted.

The HST will hold a total of 600 ml of solvent. However, through trial and error it has been found that at -80°C , using methyl alcohol, 150 ml solvent is an adequate amount. In putting the HST together it was found helpful to liberally apply silicon grease around the bottom where the ring is seated.

HST must be kept at -80°C must be done with a dry ice/acetone slush.

SIGNATURE AND DATE

READ AND UNDERSTOOD

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